

Guitar Pickup Winder - Mechanical Design Guide

Complete Build Strategy for Professional Results

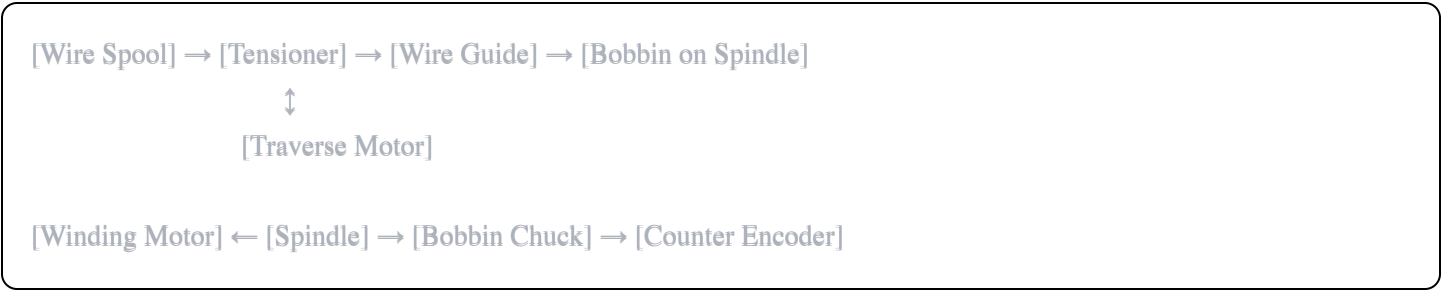
Design Philosophy

A good pickup winder needs:

- ✔ **Precision** - Accurate turn counting and wire placement
- ✔ **Consistency** - Repeatable results
- ✔ **Speed** - Reasonable winding times (5-15 minutes)
- ✔ **Tension Control** - Even wire tension throughout
- ✔ **Rigidity** - No flex or vibration

Overall Layout

Recommended Configuration:



Key Distances:

- Tensioner to bobbin:** 200-400mm (adjustable)
- Wire guide to bobbin:** 10-30mm (close for accuracy)
- Base size:** Minimum 400mm x 300mm for stability

Material Recommendations:

- Frame:** Aluminum extrusion (20x20 or 40x40) or plywood (15mm+)
- Mounting plates:** Aluminum or steel for rigidity
- Spindle:** Hardened steel rod or precision shaft
- Bearings:** Quality sealed ball bearings (608 or 6001)

Winding Spindle Assembly

Option A: Direct Drive (Simplest)

[LDO Motor] → [Flexible Coupler] → [Spindle Shaft] → [Chuck]
(5mm to 8mm) (8mm steel) (Holds bobbin)

Components:

- **Motor shaft:** 5mm (LDO standard)
- **Coupler:** 5mm to 8mm flexible or rigid
- **Shaft:** 8mm hardened steel, 150-200mm long
- **Bearings:** 2x 608ZZ or 6001ZZ
- **Chuck:** 3-jaw chuck, ER11 collet, or custom bobbin holder

Option B: Belt Drive (Better for high speed)

[LDO Motor] → [Pulley] ~~Belt~~ [Pulley] → [Spindle]
(16T) (48T) (Reduction)

Advantages:

- Quieter operation
- Belt acts as shock absorber
- Easier to change gear ratios
- Motor can be mounted remotely

Belt options:

- GT2 timing belt (2mm pitch)
- 6mm or 9mm width

Critical Spindle Requirements:

1. **Concentricity:** <0.1mm runout
2. **Bearing support:** Minimum 50mm apart
3. **Overhang:** Keep bobbin close to front bearing
4. **Balance:** Use counterweight if heavy bobbins

Bobbin Chuck Design

Option 1: Collet Chuck (Professional)

Uses ER11 or ER16 collet system

- Holds 1-10mm arbors
- Quick change
- Excellent concentricity
- Cost: \$20-40

Option 2: 3-Jaw Chuck (Versatile)

Mini lathe chuck (50mm)

- Adjustable grip
- Holds various shapes
- Needs adapter to spindle
- Cost: \$15-30

Option 3: Custom Arbor (Cheapest)

Precision-turned arbor for specific bobbin

- Snug fit on bobbin center hole
- Set screw to lock
- Make several for different sizes
- Cost: DIY or \$5-10 machined

Critical: Whatever you use, ensure the bobbin is:

- Centered (check with dial indicator if possible)
- Secure (won't slip during high-speed winding)
- Perpendicular to spindle axis

Traverse System Integration

Mounting Your Linear Slide:

[Base Plate]



[Linear Slide] ← Your 100mm slide



[Wire Guide] ← Moves back and forth

Positioning:

1. **Height:** Wire guide should be at bobbin centerline $\pm 10\text{mm}$
2. **Distance from bobbin:** 10-30mm (closer = more accurate)
3. **Parallel:** Slide must run parallel to spindle axis
4. **Level:** Check with spirit level or dial indicator

Wire Guide Design:

Simple Eye Guide:



Traverse carriage

Eye guide specifications:

- **Material:** Hardened steel, ceramic, or ruby insert
- **Hole size:** 0.5-1.0mm for AWG 42-44 wire
- **Finish:** Polished smooth (no scratches!)
- **Shape:** Rounded edges to prevent wire damage

Advanced: Use jewel orifice (watch bearing jewel) for best results

Traverse Synchronization:

For proper winding, traverse must coordinate with winding speed:

Traverse pitch = Wire diameter $\times$ winds per layer

Example for AWG 42 wire (0.064mm):

- Single layer: Move 0.064mm per wind
- 90% fill: Move 0.058mm per wind

In firmware:

- Calculate steps needed per wind
 - Move traverse incrementally
 - Reverse at bobbin edges
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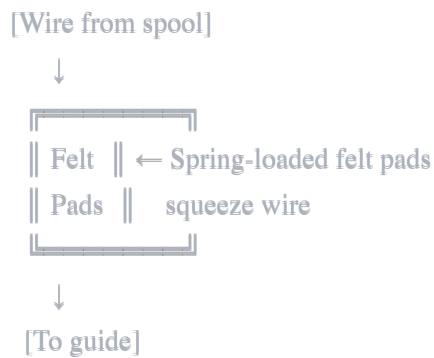
Wire Tensioning System

Why Tension Matters:

-  **Too loose:** Sloppy winding, air gaps, poor tone
-  **Too tight:** Wire breaks, deformed bobbin
-  **Just right:** Consistent turns, good tone, reliable

Target tension for AWG 42-44: 20-40 grams force

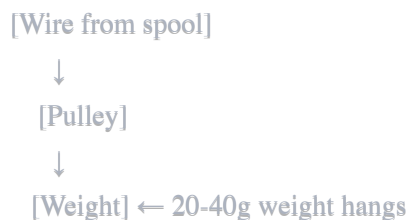
Option 1: Felt Pad Tensioner (Simplest)



Build:

- 2 felt pads with spring pressure
- Adjustable screw to set tension
- Wire passes between pads

Option 2: Weight Tensioner (Most consistent)



Advantages:

- Constant tension (gravity)
- Simple, reliable
- Easy to adjust (change weight)

Option 3: Magnetic Tensioner (Advanced)

Permanent magnet creates drag

- Infinitely adjustable
- Very consistent
- Requires precision setup

Pro tip: Combine methods - weight for consistency + felt for damping

Encoder Mounting

Spindle-Mounted (Most Accurate)

Mount encoder directly on spindle shaft:

[Motor] → [Shaft] → [Encoder Disk] → [Bobbin]
↓
[Sensor]

Your encoder (360 PPR):

- Mount disk on shaft
- Position sensor close (1-3mm gap)
- Ensure sensor alignment
- Secure rigidly (no wobble!)

Critical Considerations:

1. **Alignment:** Sensor must be perpendicular to disk
2. **Gap:** Follow encoder specs (usually 1-3mm)
3. **Rigidity:** Sensor mount must NOT vibrate
4. **Cable routing:** Keep encoder cable away from motor wires (noise!)

Z-Index Usage:

Your encoder has index (Z) pulse:

- Triggers once per revolution
 - Use to verify count accuracy
 - Can reset counter to "0" each turn
 - Useful for multi-layer windings
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Assembly Best Practices

Step 1: Build Rigid Base

Use:

- Heavy plywood (15mm+)
- Aluminum extrusion frame
- Steel plate

Add:

- Rubber feet to dampen vibration
- Weight for stability

Step 2: Mount Motor & Spindle

Order:

1. Mount motor securely
2. Install spindle with bearings
3. Align motor to spindle (≤ 0.5 mm offset)
4. Connect coupler or belt
5. Check for smooth rotation (no binding)

Alignment check:

- Spin by hand - should be smooth
- No wobble visible
- Minimal noise when powered

Step 3: Mount Traverse

1. Position slide parallel to spindle

2. Bolt down securely
3. Install wire guide on carriage
4. Adjust height to bobbin center
5. Test full travel (no binding)

Step 4: Install Tensioner

1. Mount between spool and wire guide
2. Position for straight wire path
3. Set initial tension (start light)
4. Test with scrap wire

Step 5: Wire Routing

Path should be:

Spool → Tensioner → Wire guide → Bobbin

Avoid:

- Sharp bends
- Contact with moving parts
- Crossing motor/signal cables

Electrical Integration

Controller Mounting:

Mount SKR Pico in enclosure:

- Adequate ventilation (drivers get warm)
- Away from wire path
- Easy access to USB port
- Display visible from operator position

Motor Cable Routing:

Keep separate from:

- Encoder signals (noise interference)
- USB cable
- Display/I2C cables

Use:

- Shielded cable for motors
- Twisted pairs for encoder
- Separate cable channels

Emergency Stop:

Mount big red button:

- Within easy reach
- Cuts power to motors
- Wired to MCP23017 BTN_ESTOP
- Test before first wind!

Calibration & Testing

1. Encoder Calibration:

Test procedure:

1. Mark bobbin with tape
2. Wind exactly 100 turns (manual count)
3. Check encoder reading
4. Should read: $100 \times 720 = 72,000$ counts
5. Adjust firmware if needed

2. Traverse Pitch Calibration:

1. Install empty bobbin
2. Wind single layer slowly
3. Measure wire spacing with caliper
4. Adjust TRAVERSE_PITCH in firmware
5. Repeat until spacing = wire diameter

3. Tension Calibration:

1. Start with light tension
2. Wind test coil
3. Check for:
 - Consistent turns
 - No loose windings

- No wire breaks
4. Adjust up/down as needed
 5. Document final setting

4. Speed Testing:

Test speeds:

- SLOW: 800 steps/sec
- MED: 2000 steps/sec
- FAST: 4000 steps/sec
- MAX: 6000 steps/sec

Check for:

- Smooth operation
- No vibration
- Motor temperature (should be warm, not hot)
- Wire tension consistency

Troubleshooting Common Issues

Problem: Inconsistent Wind Count

Causes:

- Encoder not secure
- Sensor gap too large
- Electrical noise
- Loose coupler

Solutions:

- Check encoder mounting rigidity
- Adjust sensor gap to spec
- Route cables properly
- Tighten all connections

Problem: Wire Breaks During Winding

Causes:

- Tension too high
- Sharp edge on wire guide
- Wire twisted or kinked
- Speed too fast

Solutions:

- Reduce tension
- Polish all wire contact points
- Check wire spool rotation
- Slow down winding speed

Problem: Sloppy or Uneven Winding

Causes:

- Tension too low
- Traverse not synchronized
- Bobbin wobble
- Wire guide too far from bobbin

Solutions:

- Increase tension slightly
- Calibrate traverse pitch
- Check spindle alignment
- Move guide closer

Problem: Motor Overheating

Causes:

- Current set too high
- Poor ventilation
- Binding in mechanism
- Continuous high-speed operation

Solutions:

- Reduce current setting in firmware

- Add cooling fan
- Check for mechanical binding
- Add pauses between winds

Material & Parts List

Mechanical Components:

Item	Qty	Est. Cost
8mm steel shaft (200mm)	1	\$5-10
608ZZ bearings	2	\$2-5
5mm to 8mm coupler	1	\$3-8
ER11 collet chuck	1	\$20-40
Aluminum plate (various)	-	\$15-30
Bearing blocks/mounts	2	\$5-15
Fasteners (M3, M4, M5)	Set	\$5-10
Wire guide eye (jewel)	1	\$5-15
Felt pads	2	\$2-5
Tension adjustment screw	1	\$2-5

Total mechanical: ~\$70-150 (depending on options)

Wire & Supplies:

- AWG 42-44 pickup wire: \$20-40/lb
- Bobbins (Strat, Tele, HB): \$1-3 each
- Tension springs: \$2-5
- Misc hardware: \$10-20

Pro Tips

For Best Results:

1. **Start slow** - Perfect technique at low speed first
2. **Document everything** - Settings, tension, speeds for each pickup type
3. **Use quality wire** - Don't skimp, cheap wire breaks easily

4. **Keep it clean** - Dust and debris affect wire quality
5. **Calibrate often** - Check encoder accuracy regularly

Common Mistakes to Avoid:

- ❌ Rushing the build (take time to align properly)
- ❌ Using cheap bearings (causes wobble)
- ❌ Ignoring wire path smoothness
- ❌ Setting tension by feel (use scale!)
- ❌ Winding too fast initially
- ❌ Not testing with scrap wire first

Advanced Techniques:

- **Scatter winding:** Randomize traverse for vintage tone
 - **Multi-layer:** Program automatic layer transitions
 - **Auto-stop:** Use Z-index to stop at exact count
 - **Data logging:** Record settings for each pickup type
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Example Build Sequence

Day 1: Frame & Spindle

1. Cut and assemble base
2. Mount motor bracket
3. Install spindle with bearings
4. Connect motor with coupler
5. Test rotation

Day 2: Traverse & Guide

1. Mount linear slide
2. Install wire guide
3. Align to spindle
4. Test full travel
5. Adjust height

Day 3: Electronics

1. Mount SKR Pico in enclosure
2. Wire motors to board
3. Install encoder
4. Connect display & buttons
5. Upload firmware

Day 4: Tensioner & Accessories

1. Build tensioner mechanism
2. Mount wire spool holder
3. Route wire path
4. Install emergency stop
5. Add safety guards

Day 5: Calibration

1. Encoder count test
2. Traverse pitch calibration
3. Tension adjustment
4. Speed testing
5. First real pickup wind!

Recommended Upgrades

Once basic winder works:

1. **Foot pedal** - Start/stop hands-free
 2. **Auto-traverse direction** - Smooth layer transitions
 3. **Preset library** - Save favorite settings
 4. **Data logging** - Track all winds to SD card
 5. **Webcam mount** - Record winding process
 6. **LED lighting** - Illuminate work area
 7. **Dust cover** - Protect when not in use
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Critical Dimensions Summary

Measurement	Target	Tolerance
Spindle runout	0mm	±0.1mm
Traverse parallel	Perfect	±0.2mm
Wire guide gap	15mm	±5mm
Encoder sensor gap	Per spec	±0.5mm
Motor alignment	Centered	±0.5mm

You're now ready to build a professional-grade pickup winder! 🎸

Take your time, measure twice, cut once, and you'll have a machine that produces pickups as good as (or better than) commercial winders!